Medical Insurance Cost Prediction using Linear Regression

Machine Learning Assignment | Regression using Linear Regression
Dataset: Medical Cost Personal Dataset (Kaggle)

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1. Introduction

Health insurance pricing depends heavily on accurately estimating an individual's expected medical costs. This report documents a complete machine learning pipeline — from data understanding through model deploy/improvement decisions — built to predict individual medical insurance charges from six demographic and lifestyle attributes. Linear Regression is used as the primary modeling technique, with Ridge, Lasso, and Polynomial Regression evaluated as improvements over the baseline.

2. Problem Statement

Health insurance companies need to estimate the expected medical expenses of a customer before issuing a policy, so that premiums are priced appropriately, financial risk is managed, and personalized plans can be designed. The objective of this project is to develop a regression model that predicts the continuous target variable **charges** using a customer's **age, sex, BMI, number of children, smoking status,** and **region** of residence.

3. Dataset Description

The dataset used is the publicly available **Medical Cost Personal Dataset** (Kaggle: mirichoi0218/insurance), containing **1,338 records** and **7 columns**: 6 input features and the target variable **charges**.

Column	Type	Description
age	Numerical	Age of primary policy beneficiary (18-64)
sex	Categorical	Biological sex (male / female)
bmi	Numerical	Body Mass Index (kg/m^2)
children	Numerical	Number of dependents covered (0-5)
smoker	Categorical	Smoking status (yes / no)
region	Categorical	Residential region (4 US regions)
charges	Numerical (Target)	Individual medical insurance cost billed

Dataset summary: No missing values were found in any column. Numerical features are age, bmi, children (and the target charges); categorical features are sex, smoker, and region.

- **Mean charge is ~$13,270** while the median is ~$9,382 — the target is strongly right-skewed, with a long tail of high-cost customers.
- **Age** is roughly uniformly distributed between 18 and 64 (mean ~39.2), representing a broad working-age population.
- **BMI** is approximately normally distributed around a mean of ~30.7, which falls in the obese range by WHO classification for the average customer.
- **Children** is heavily right-skewed toward 0 — most policyholders have no dependents on the plan.
- **Sex and region are both close to evenly split** across their categories, so there is no meaningful class-imbalance concern for these variables.

4. Exploratory Data Analysis

4.1 Target Variable Distribution

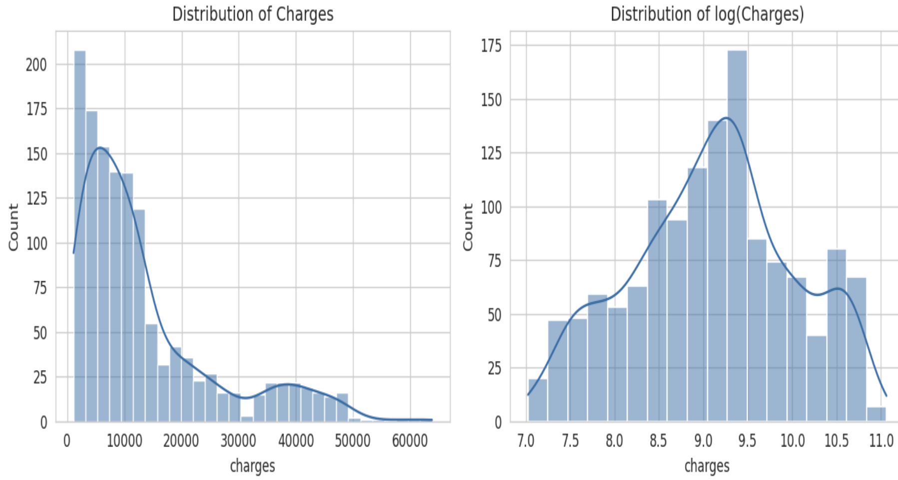

Figure 1: Distribution of raw charges (left, right-skewed) vs. log-transformed charges (right, closer to normal).

The raw **charges** distribution is heavily right-skewed with a long tail of high-cost customers; a log transform makes the distribution noticeably more symmetric, which is useful context even though the model in this report is trained on the raw target for direct dollar interpretability.

4.2 Correlation Analysis

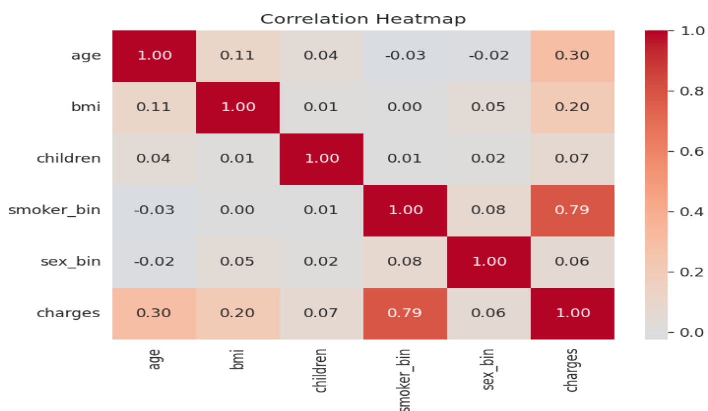

Figure 2: Correlation heatmap of numeric and encoded categorical features against charges.

Smoking status has by far the strongest correlation with charges (r ≈ 0.79), followed by age (r ≈ 0.30) and BMI (r ≈ 0.20). Children and sex show weak correlation with the target (r < 0.10).

4.3 Feature Distributions

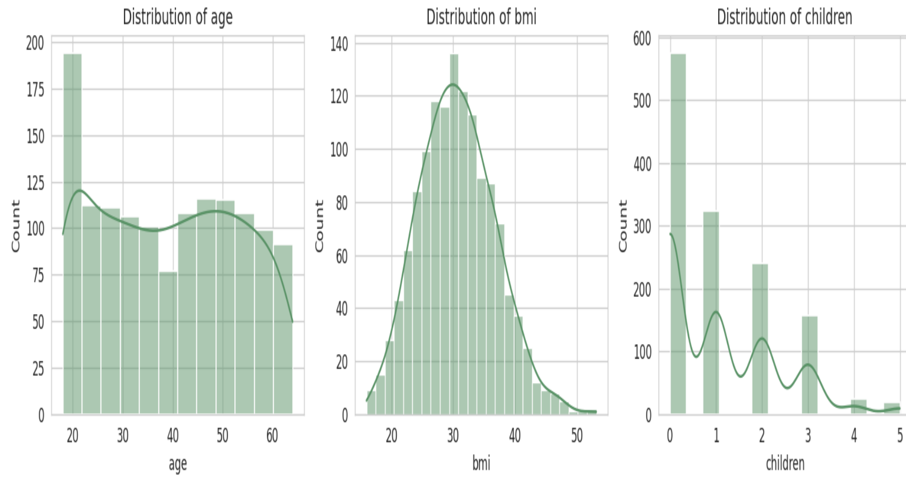

Figure 3: Distributions of age, BMI, and number of children.

4.4 Charges by Categorical Groups

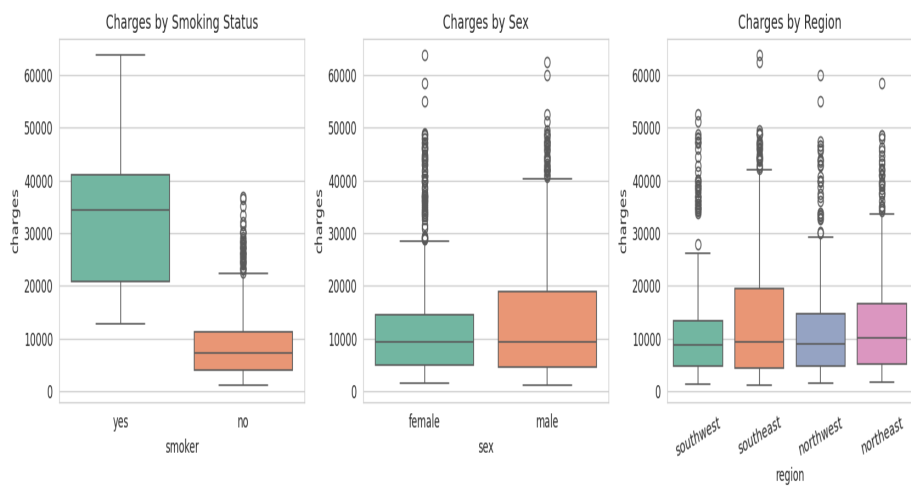

Figure 4: Charges distribution by smoking status, sex, and region.

Smokers show a dramatically higher median charge and wider spread than non-smokers. Sex and region show comparatively minor differences in the distribution of charges.

4.5 Relationships Between Features and Charges

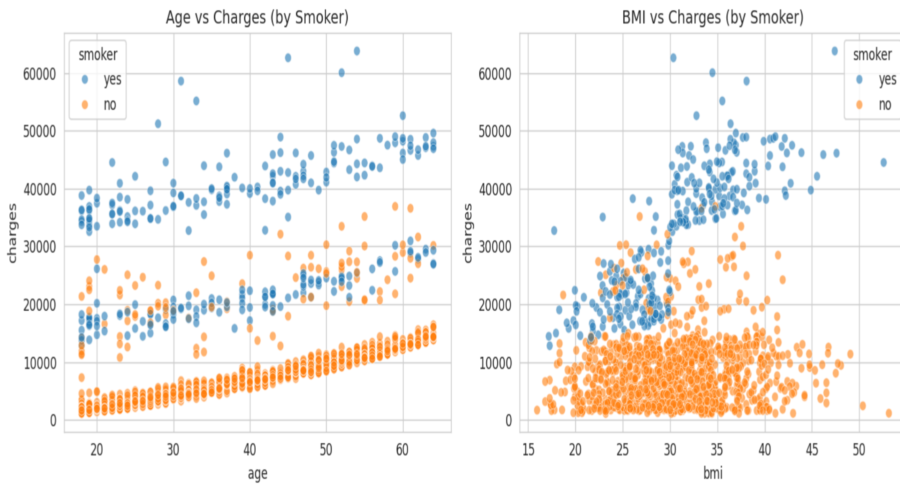

Figure 5: Age vs. charges and BMI vs. charges, colored by smoking status.

Both scatter plots reveal **two distinct clusters** separated almost entirely by smoking status: within each cluster, age and BMI still show a mild positive trend with charges, but the smoker/non-smoker split dominates the overall pattern — and for smokers, higher BMI corresponds to a much steeper rise in charges than for non-smokers, indicating a real interaction effect.

4.6 Pairwise Relationships

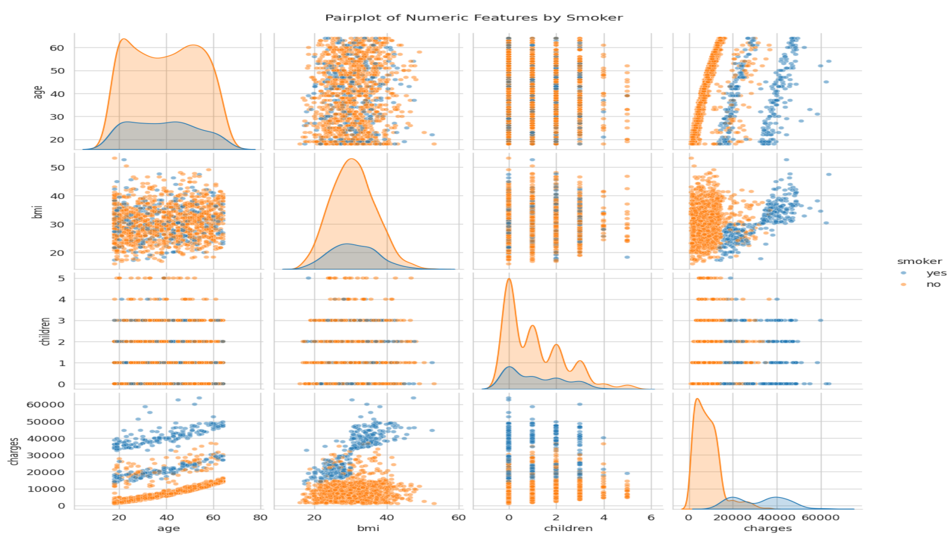

Figure 6: Pairplot of age, BMI, children, and charges, colored by smoking status.

4.7 Outlier Analysis

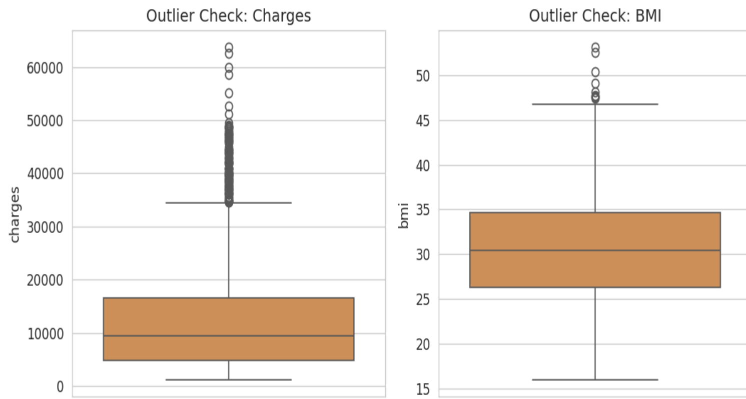

Figure 7: Boxplots highlighting outliers in charges (139 by IQR rule) and BMI (9 by IQR rule).

The IQR method flags 139 outliers in charges (mostly high-BMI smokers) and 9 in BMI. These are genuine, naturally occurring high-cost/high-BMI cases rather than data-entry errors, so they were retained in modeling — a usable pricing model must be able to handle exactly this kind of high-risk customer.

EDA Summary Observations

- **Strongest correlated feature with charges:** smoker ($r \approx 0.79$), by a wide margin over age and BMI.
- **Outliers exist** primarily in charges, concentrated among smokers with elevated BMI — these represent real high-risk customers, not errors.
- **Most useful predictors:** smoker, age, and BMI, especially the interaction between smoker status and BMI/age.

5. Data Preprocessing

- **Missing values:** none were found, so no imputation was required.
- **Label Encoding:** the binary categorical variables sex (male=1/female=0) and smoker (yes=1/no=0) were label-encoded, since a single 0/1 column fully captures a two-level category with no artificial ordering.
- **One-Hot Encoding:** the four-level region variable was one-hot encoded (with the first category dropped to avoid the dummy variable trap / multicollinearity), since region has no natural order and label-encoding it would incorrectly imply one.
- **Feature Scaling:** all features were standardized with StandardScaler (zero mean, unit variance). This is not strictly required for plain Linear Regression, but it is necessary for a fair comparison of coefficient magnitudes, for Ridge/Lasso regularization to penalize features evenly, and for numerically stable polynomial feature expansion.
- **Train-Test Split (80:20):** data was split with a fixed random seed, so that model performance is honestly evaluated on customers the model has not seen during training — this is what determines real-world usefulness.

6. Feature Engineering

Beyond standard encoding and scaling, the key feature engineering decision in this project was the use of **degree-2 polynomial feature expansion** as a model-improvement step (Part 6 of the assignment / Section 8 of this report). This automatically generates squared terms and pairwise interaction terms (e.g. smoker × bmi, age × bmi), letting a linear model approximate the non-linear interaction between smoking status, age, and BMI that a first-order linear model cannot represent on its own.

7. Model Building

A baseline **Linear Regression** model (scikit-learn) was trained on the scaled, encoded features to predict charges.

Model Coefficients

Feature	Coefficient
smoker	9,558.48
age	3,614.98
bmi	2,036.23
children	516.89
region_southwest	-349.11
region_southeast	-290.16
region_northwest	-158.14
sex	-9.29

Intercept: 13,346.09 (predicted charges for a customer at the average value of every standardized feature).

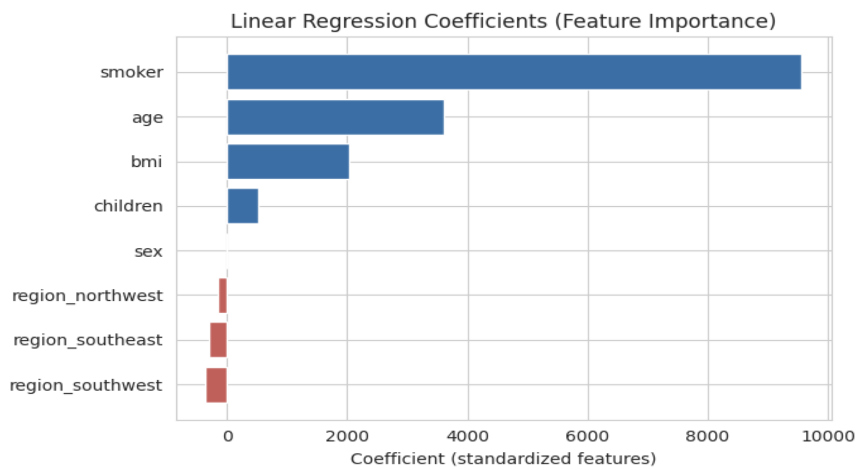

Figure 8: Linear Regression coefficients (standardized features), showing relative feature importance.

Interpretation

- **Positive coefficients** (smoker, age, bmi, children) indicate that increases in these values are associated with higher predicted charges — smoker has by far the largest effect.
- **Negative coefficients** (region dummies, sex) are small in magnitude, indicating these features have little practical influence on predicted charges relative to the baseline region and to females respectively.
- **Overall relationship:** smoking status is the dominant driver of predicted cost, with age and BMI as secondary but genuine risk factors; sex, children, and region contribute comparatively little explanatory power.

8. Model Evaluation

Metric	Value
MAE	4,181.19

MSE	33,596,915.85
RMSE	5,796.28
R2 Score	0.7836
Adjusted R2	0.7769

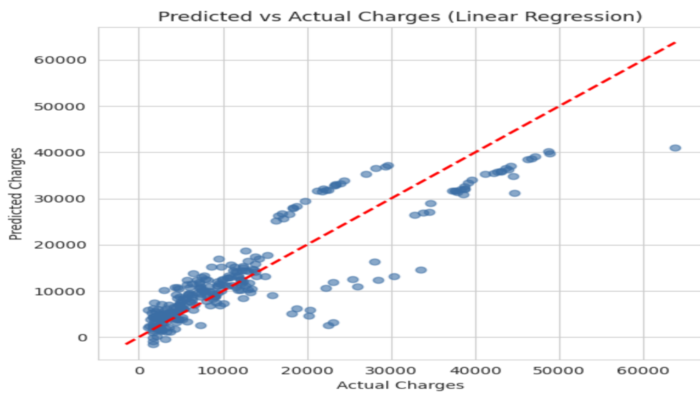

Figure 9: Predicted vs. actual charges on the held-out test set (baseline Linear Regression).

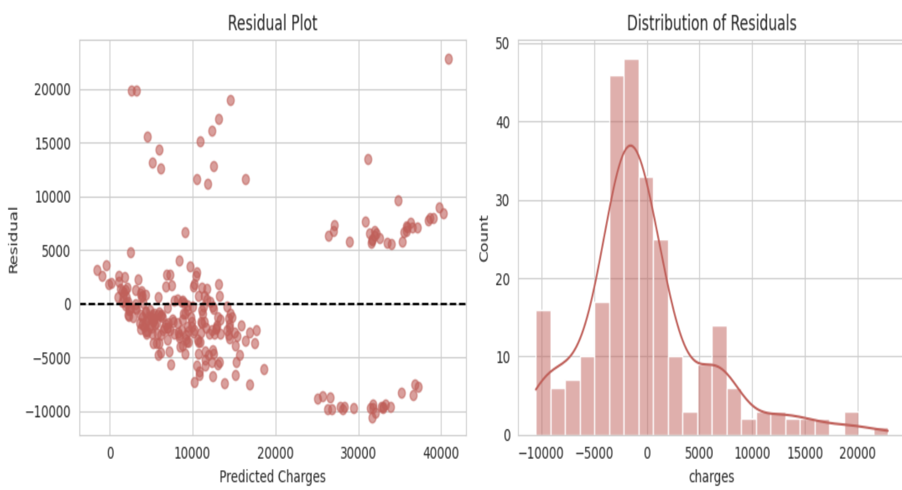

Figure 10: Residual plot and residual distribution for the baseline model.

The baseline Linear Regression model explains about **78%** of the variance in charges (R^2 = 0.7836) on unseen test data. The residual plot shows a **funnel/fan pattern** — residual spread increases with predicted charges rather than being uniformly random — indicating the plain linear model is mildly misspecified, most likely because it can't represent the non-linear smoker × age/BMI interaction visible in the EDA.

Most Suitable Evaluation Metric

RMSE is the most practically suitable metric for this business problem: it is denominated in dollars (directly interpretable) and penalizes large errors more heavily than MAE, which matters because under-predicting a genuinely high-cost customer is a more expensive mistake for an insurer than a small error on a low-cost customer. **R^2/Adjusted R^2** is best for judging overall explained variance and for fairly comparing models of different complexity (Adjusted R^2 penalizes added features that don't improve fit). In practice these metrics are reported together rather than any single one being used in isolation.

9. Model Improvement / Hyperparameter Tuning

Three improvement strategies were evaluated against the baseline Linear Regression model: **Ridge Regression** and **Lasso Regression** (both hyperparameter-tuned over alpha via 5-fold GridSearchCV), and **Polynomial Regression (degree 2)**.

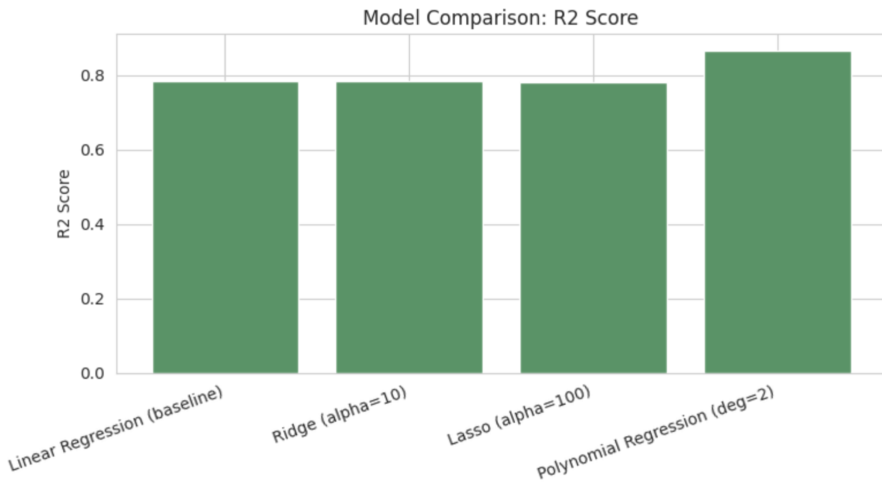

Figure 11: R2 score comparison across baseline and improved models.

Model	MAE	RMSE	R2	Adj. R2
Linear Regression (baseline)	4,181.19	5,796.28	0.7836	0.7769
Ridge (alpha=10)	4,197.66	5,803.95	0.7830	0.7763
Lasso (alpha=100)	4,210.61	5,835.80	0.7806	0.7739
Polynomial Regression (deg=2)	2,729.50	4,551.13	0.8666	0.8403

- Ridge and Lasso perform almost identically to the baseline — expected, since with only 8 features and no severe multicollinearity, regularization has little room to improve on plain OLS.
- **Polynomial Regression (degree 2) is the clear winner:** R^2 improves from 0.78 to 0.87 and RMSE drops from ~5,796 to ~4,551, because it captures the non-linear smoker-age-BMI interaction that a linear model cannot represent.
- **Trade-off:** the polynomial model is more complex and less directly interpretable than plain linear coefficients, so a production deployment would weigh this accuracy gain against interpretability and overfitting-risk considerations.

10. Business Insights

1. Which factors have the greatest impact on medical insurance charges?

Smoking status, by a wide margin, followed by age and BMI. Sex, number of children, and region have comparatively minor impact.

2. Does smoking significantly increase insurance costs?

Yes, dramatically: smokers pay ~$32,050 on average vs. ~$8,434 for non-smokers (~3.8x higher), and smoker status has the single largest model coefficient (+9,558).

3. How does BMI affect insurance expenses?

Higher BMI is associated with higher charges (coefficient +2,036), and this effect is much steeper specifically among smokers — the polynomial model's improvement over the linear model largely comes from capturing this smoker-BMI interaction.

4. Does age influence insurance charges?

Yes — age has the second-largest positive coefficient (+3,615) and a moderate positive correlation with charges ($r \approx 0.30$), consistent with higher health risk at older ages.

5. Which feature contributes the most to predicting insurance charges?

Smoker status contributes the most, both by raw correlation ($r \approx 0.79$) and by coefficient magnitude in the trained model.

6. How can insurance companies use these predictions for premium pricing and risk assessment?

- Price premiums using risk tiers primarily driven by smoking status, then age and BMI, since these are the features with genuine, large, and consistent predictive power.
- Flag the smoker + high-BMI segment for more careful underwriting/risk review, since it is the highest-cost segment by a wide margin.
- Avoid over-engineering pricing around sex, children, or region, since these add little predictive value and mainly add administrative complexity.
- Use the more accurate polynomial model internally for risk scoring/premium calculation, while using the simpler linear coefficients to explain and justify pricing decisions to regulators and customers.

Customer Segment Analysis

Segment identified as highest-cost: **smokers with BMI ≥ 30** (obese smokers). This segment's average charges are roughly **4x** higher than the rest of the population, confirming that smoking and BMI interact rather than acting independently.

Strategies to Reduce Healthcare Costs

- Offer premium discounts or structured incentive programs for smoking cessation — the single highest-leverage lifestyle factor identified.
- Tie wellness incentives (gym memberships, health coaching, BMI-reduction programs) specifically to the smoker + high-BMI segment, where the payoff is largest.
- Use age-banded premiums so pricing scales appropriately with age-related risk without over-penalizing younger, lower-risk customers.

11. Conclusion

A Linear Regression model using age, sex, BMI, children, smoker, and region explains roughly **78%** of the variance in medical insurance charges ($R^2 \approx 0.7836$), with **smoking status** emerging as by far the strongest single predictor, followed by age and BMI. Because charges depend on a genuine non-linear interaction between smoking, age, and BMI, a **degree-2 polynomial regression** captures this relationship better, improving R^2 to roughly **0.87** with a meaningfully lower RMSE, at some cost to interpretability. For a health insurer, a practical approach is to use the interpretable linear coefficients to communicate and justify pricing tiers to regulators and customers, while using the higher-accuracy polynomial model internally for actual premium calculation and risk scoring.